

Netflix Report

1. Loading data and cleaning

First, I filtered and cleaned the title data. I filtered the data to only include movies. I cleaned the data to remove column formatting issues. As some of the columns had two values per observation (e.g. multiple genres), I pivoted longer to allow for genre and country based analysis.

2. Types of movies by country

I first identify the identify the top countries. The five countries which are the most prolific producers are the United States, India, France, and Canada.

Table 2.1: Top 5 Genres by Country

production_countries	genres	n
US	comedy	630
US	drama	618
US	documentation	365
US	thriller	353
US	action	251
IN	drama	448
IN	comedy	252
IN	romance	192
IN	thriller	174
IN	action	149
GB	drama	116
GB	thriller	80
GB	comedy	77
GB	documentation	59
GB	action	49
FR	drama	113
FR	european	100

production_countries	genres	n
FR	comedy	51
FR	thriller	51
FR	crime	36
CA	drama	63
CA	comedy	56
CA	thriller	40
CA	animation	31
CA	family	31

The results from analysis the top genres by countries illustrate heterogeneity in preferences. Comedy and drama dominate in the United States, India, and Canada. Documentation films are strong in the United States, and the UK.

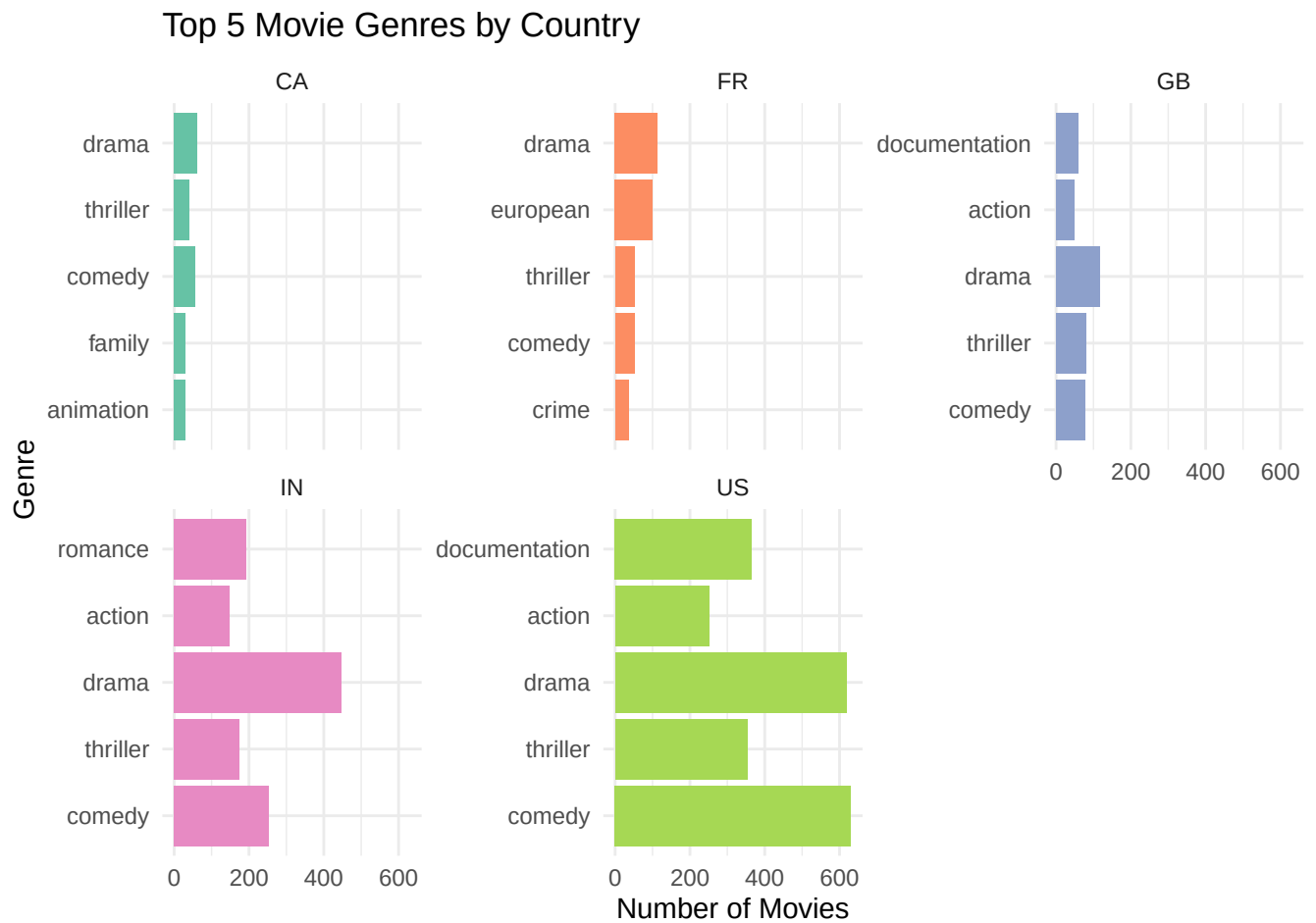


Figure 2.1: Top Generes by Country.

3. Distribution of ratings by genre

Here, I visualise the distribution of ratings by genre. History, documentation, war and sport films are more commonly well-received. Horror, family, sci-fi, and action lie at the bottom range of the rating distribution. However, most genres cluster around a mean score of between 6 and 7, and are fairly widely spread. This indicates genre alone is a weak predictor of ratings.

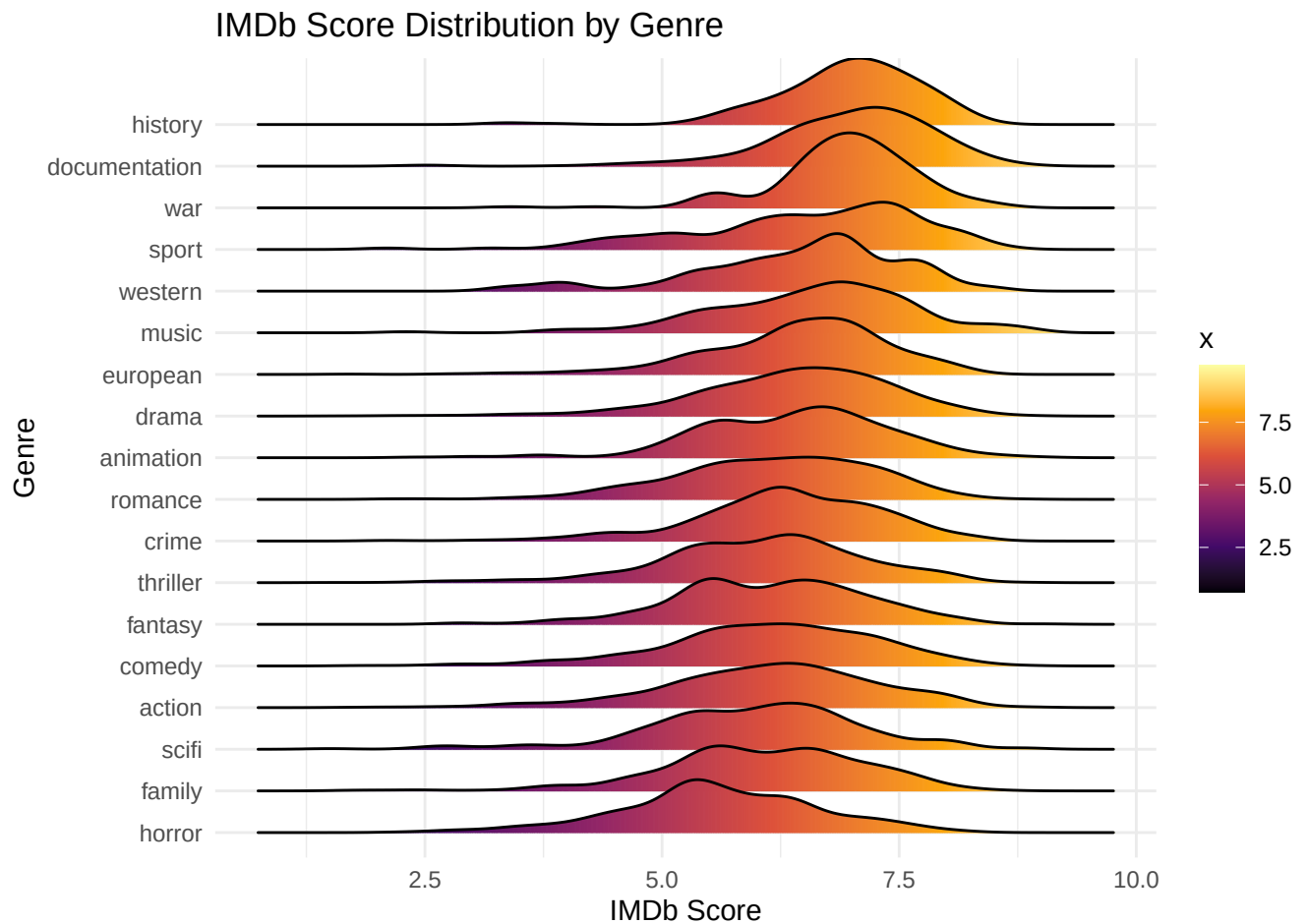


Figure 3.1: IMDb Score Distribution by Genre.

4. Distribution of runtime by genre

Here, I visualise the distribution of runtime by genre. Similar to the ratings distribution, most genres are clustered over a similar mean (just below 100 minutes) with fairly wide spreads.

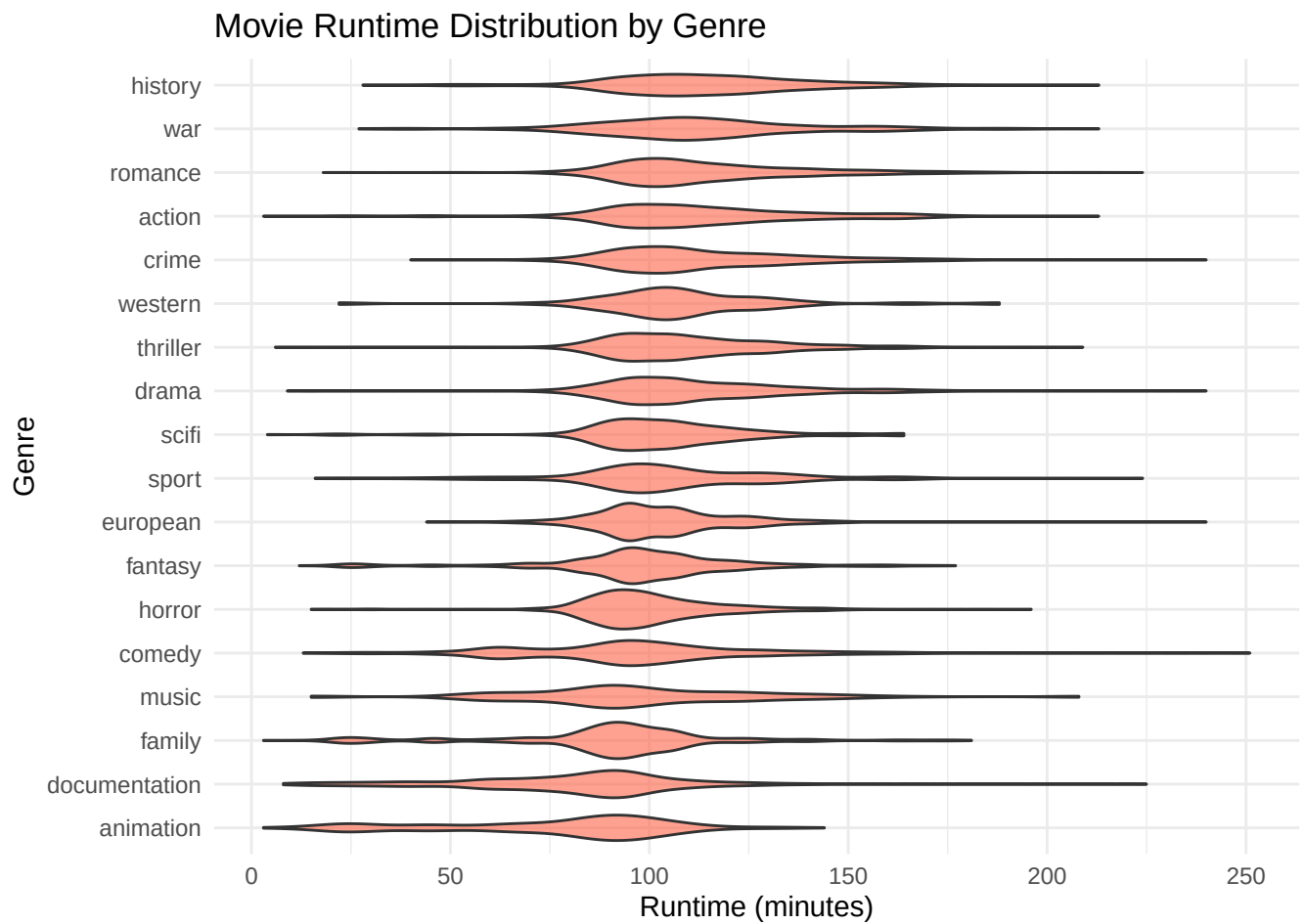


Figure 4.1: Movie Runtime Distribution by Genre.

5. Text analysis

First, I clean and filter the dataset. To avoid double counting words from movies with multiple genres, the data is reduced to one row per movie per country before tokenising descriptions.

Second, I look for the top words by country. Across most countries, “life”, “family”, and “world” are common in movie descriptions. This indicates that a strong drama/relationship focus is appealing. The UK stands out with location-specific words like “london” and “british”. Similarly, France’s results lean toward “french” and “paris”. This suggesting more nationally-rooted storytelling is favoured in these countries. In contrast, Canada and the United states have more broad themes, such as “family”, “life” and “home”.

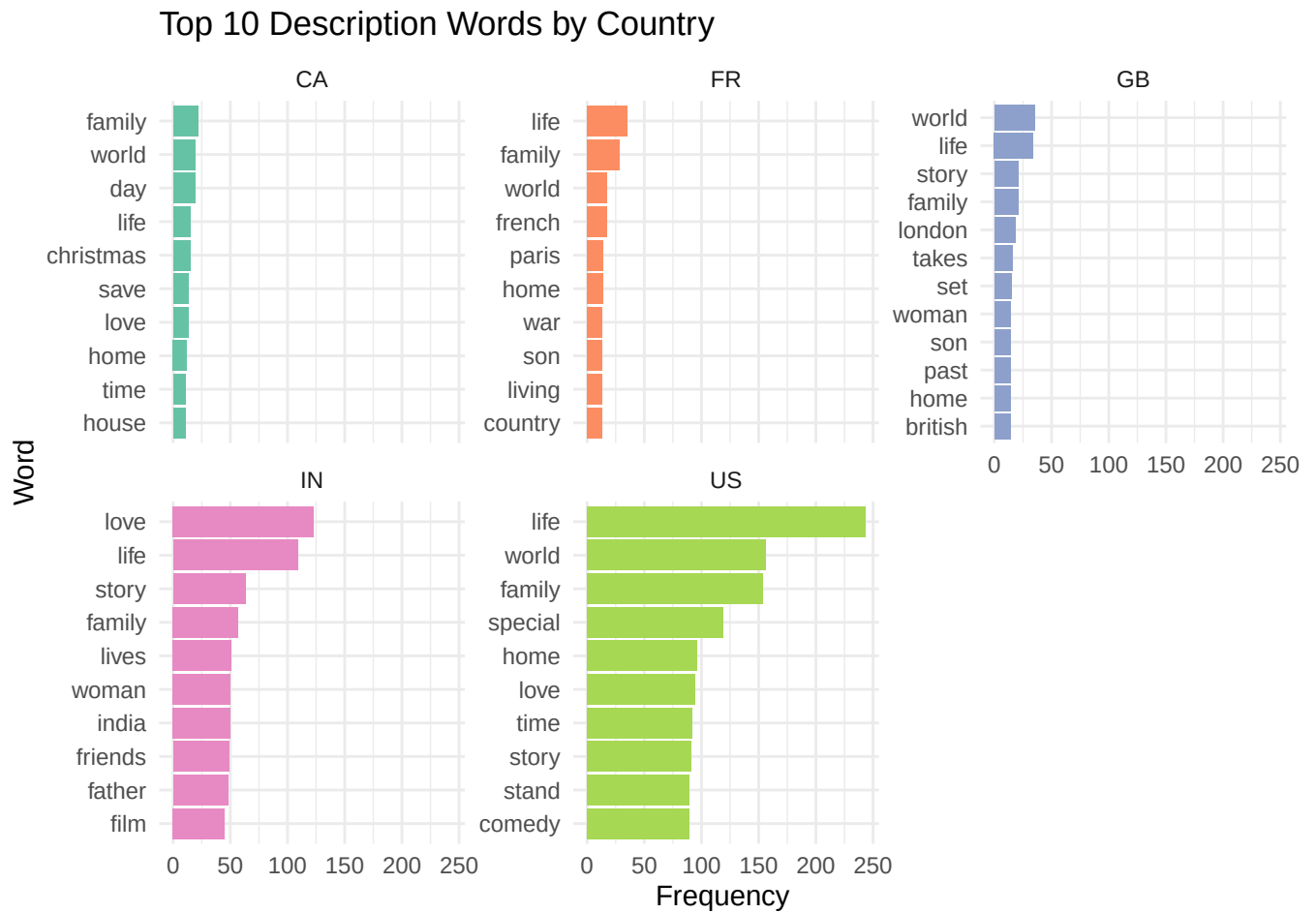


Figure 5.1: Top 10 Description Words by Country.

South Africa's most frequent words have a slightly different tilt than the world's top producers. Notably, frequent words include "south" and "africa", indicating the importance of domestic storytelling in the country. "Apartheid" is also amongst the most frequent, illustrating the importance of the country's history. Similar to the patterns observed in the rest of the world, words such as "love", "life" and "family" are also common in South Africa.

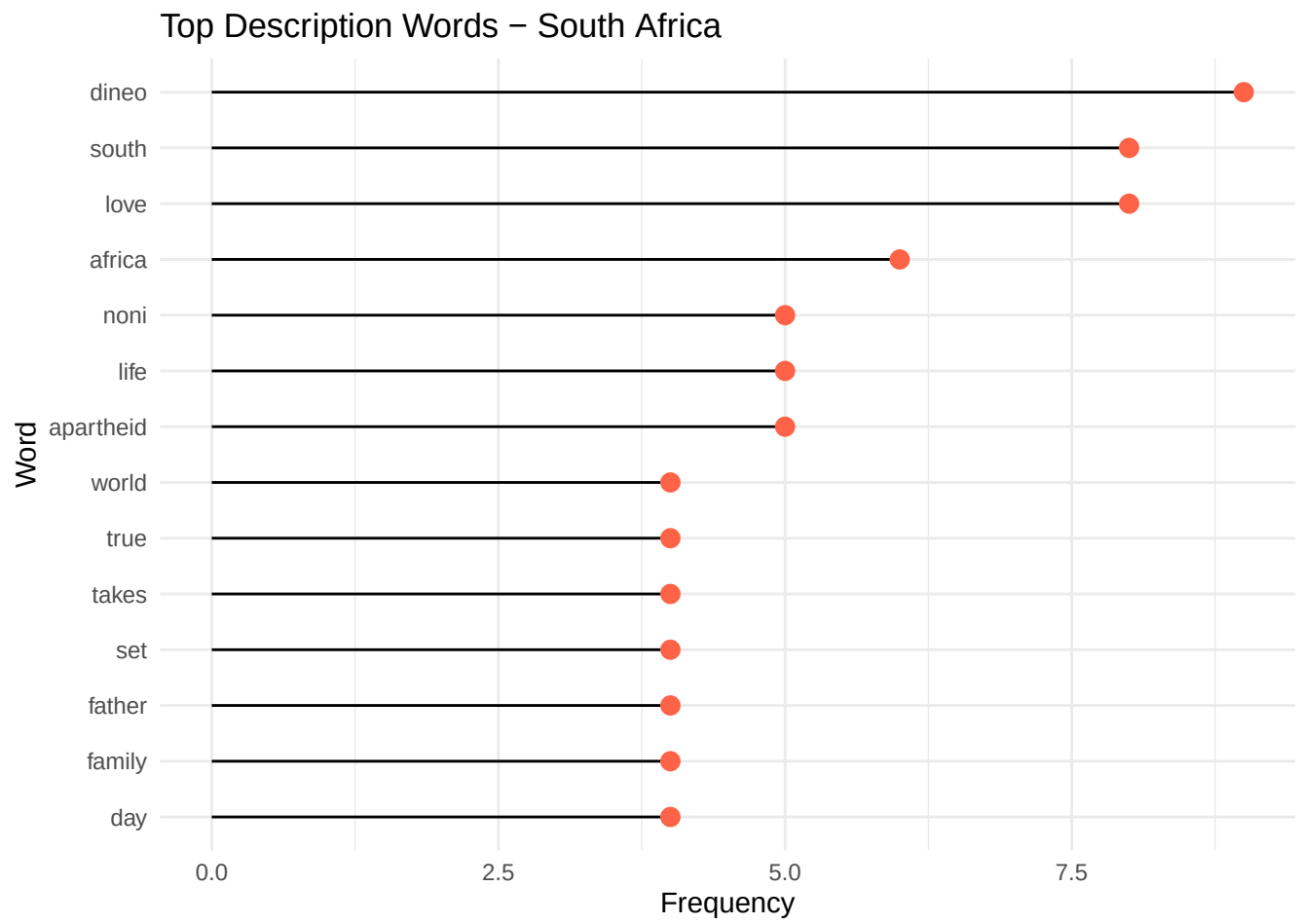


Figure 5.2: Top Description Words South Africa.